Do Visual Grounding Decoders Need Feed-Forward Networks?

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Abstract—Do feed-forward networks (FFNs) in visual grounding decoders remain necessary after a pretrained vision-language model has contextualized the image and language? We compare a four-block attention-only decoder (A4), a matched attention-plus-FFN decoder (S4), and an eight-block attention-only parameter control (A8) over frozen VLM features. A4 matches S4 on RefCOCOg and is 0.79 percentage points better on Ref-Adv-s. FineCops-Ref exposes a small A4 deficit of 0.52 points at IoU@0.5; A8 recovers it. A4 uses 44.4% fewer trainable decoder parameters and 10.1% lower cached-decoder latency. The evidence supports a bounded conclusion: FFNs are often dispensable in this small trainable grounding decoder, while the frozen VLM and its pretrained FFNs remain unchanged.

I. Introduction

Visual grounding maps an image and a referring expression to one target box. Standard Transformer grounding heads combine attention with a token-wise feed-forward network (FFN) in each block. This is a sensible default when the model must construct representations from raw inputs.

Our question is narrower. If a frozen vision-language model (VLM) already provides contextual image patches and text states, does a small task decoder still need its FFN residuals? We delete only those residuals while keeping the frozen features, projections, attention, readout, loss, optimizer, and box conversion fixed.

We compare A4, a four-block FFN-free decoder, with S4, the identical four-block decoder with FFNs. A8 doubles attention-only depth to approximately match S4's trainable parameter count. This separates a fixed-depth deletion test from a capacity-reallocation test.

The study asks three questions: (1) does A4 retain standard grounding? (2) does a released or official difficulty variable expose an FFN advantage? and (3) can additional attention depth recover a fixed-depth gap?

The answer is mixed. A4 retains standard and adversarial grounding. FineCops-Ref shows a small fixed-depth compositional gap, and A8 recovers the overall difference. No evaluated difficulty scale shows a monotonic A4 failure boundary.

II. Method

A. Frozen representation and decoder

The primary frozen backbone is SigLIP 2 Base/16 at 384 pixels. It yields 576 image tokens on a 24×24 grid and contextual text states. Linear projections map both streams to width 256.

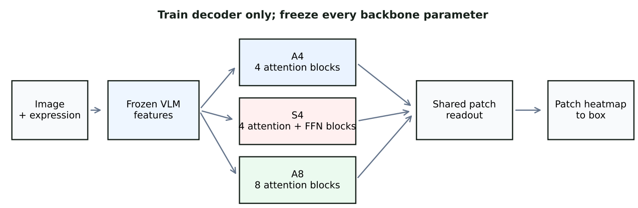

Fig. 1. The frozen VLM, projections, readout, supervision, and box conversion are shared. Only decoder capacity allocation changes.

A learned query alternates text and image cross-attention. S4 applies a pre-normalized GELU FFN after each attention pair. A4 omits these residuals. A8 uses eight attention-only blocks and has approximately the same trainable parameter count as S4.

The final query scores the 576 image patches. A softmax produces a patch distribution, trained against the target box's normalized patch-overlap distribution. A deterministic mass-to-box rule converts the heatmap to one box. There is no coordinate-regression MLP, segmentation refinement, auxiliary loss, multi-query head, or backbone adaptation.

B. Controlled comparison

The heatmap mass threshold $\tau = 0.8$ is selected once on RefCOCOg validation and frozen before held-out evaluation. All variants share data order, optimizer, learning rate, training budget, and evaluation code.

For three-seed comparisons, model predictions are paired per example. We use 10,000 image-clustered bootstrap replicates so expressions from one image are not treated as independent images. Positive differences favor A4.

III. Experimental Setup

A. Benchmarks

RefCOCOg UMD is the primary three-seed study. It supplies standard referring expressions and modality-shuffle controls. RefCOCO UNC is a seed-0 classic-dataset check.

Ref-Adv-s is an evaluation-only stress test with released negation, expression-length, and distractor metadata. FineCops-Ref is the controlled compositional test; we use its official positive-test split, levels, and tuple types without inventing post-hoc semantic labels.

TABLE I
Primary paired differences at IoU@0.5. Positive values favor A4.

Evaluation	A4–S4 (pp)	Interval
RefCOCOg direct	+0.26	90% CI $[-0.33, +0.84]$
Ref-Adv-s	+0.79	95% CI $[+0.06, +1.52]$
FineCops-Ref	-0.52	95% CI $[-0.95, -0.12]$
FineCops A8–S4	+0.26	paired estimate
CLIP control	+0.18	one seed

We also report a descriptive one-seed frozen CLIP L/14@336 control. It preserves the 24-by-24 patch geometry while changing the representation family.

B. Metrics and efficiency

The primary metric is the fraction of examples with IoU at least 0.5 (IoU@0.5). We report percentage-point differences with bootstrap intervals when available. Efficiency uses the same frozen SigLIP2 features and measures cached-decoder latency separately from full-pipeline latency.

IV. Results

A. Standard and adversarial grounding

On RefCOCOg direct expressions, A4 is 0.26 points above S4 (90% CI $[-0.33, +0.84]$). The direct-retention gate passes, while the planned direct-versus-logical interaction is not confirmed.

On Ref-Adv-s, A4 is 0.79 points above S4 (95% CI $[+0.06, +1.52]$). Negation, expression length, and distractor-count slices do not show a monotonic A4 loss. Image and text shuffles sharply reduce both A4 and S4, indicating that both modalities matter.

B. Controlled composition

FineCops-Ref reveals the only clear overall fixed-depth deficit. A4 is 0.52 points below S4 (95% CI $[-0.95, -0.12]$) over 9,605 positive-test examples.

The deficit is similar at official levels 1 and 2. Level 3 slightly favors A4 but is imprecise. Tuple-type rows are descriptive and do not establish that more relations require more FFN computation.

C. Capacity reallocation

A8 is 0.26 points above S4 on FineCops overall. Since A8 is parameter-matched rather than compute-matched, this result supports attention capacity as a substitute at the measured scale. It does not prove that FFNs never provide a distinct function.

D. Efficiency and transfer

A4 has 2.64M trainable decoder parameters versus 4.74M for S4, a 44.4% reduction. Cached-decoder latency falls from 7.46 ms to 6.71 ms, a 10.1% reduction. Full-pipeline latency changes by only 0.4% because the frozen VLM dominates.

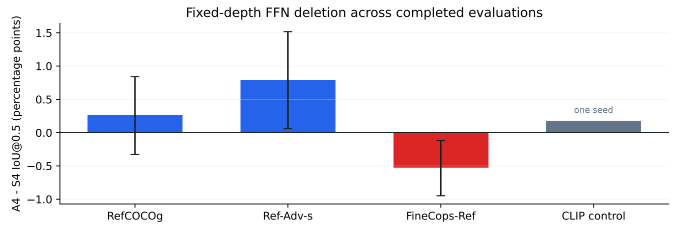

Fig. 2. A4–S4 differences across completed evaluations. FineCops-Ref is the only overall comparison with a clear fixed-depth A4 deficit.

The frozen CLIP control has A4–S4 = +0.18 points on RefCOCOg. It is one seed and descriptive only, so it supports directional transfer rather than a new significance claim.

V. Related Work and Limitations

MDETR and Grounding DINO retain standard Transformer FFNs in multimodal detection and grounding stacks [1], [2]. Frozen-CLIP region scoring and attention-based frozen-LVLM grounding show that pretrained representations contain useful localization signals [3], [4], [5]. Our intervention isolates a matched trainable decoder FFN comparison.

Attention-only Transformer studies motivate reallocating FFN capacity into depth [6]. Synthetic compositional grounding work also shows that depth and composition interact [7]. We test the corresponding capacity question on real images over frozen VLM features.

The claim is bounded to a one-query box-grounding decoder. The backbone is frozen and retains its pretrained FFNs. The output is one box from a 24-by-24 patch heatmap, not segmentation or multi-object detection. FineCops level-3 and small tuple slices are noisy. RefCOCO UNC and CLIP are descriptive replications.

VI. Conclusion

In this frozen-VLM grounding decoder, fixed-depth FFN removal preserves standard and adversarial performance. A small compositional deficit appears on FineCops-Ref, and additional attention depth recovers it at a similar parameter budget.

The supported conclusion is architectural and narrow: when a strong frozen VLM has already contextualized visual and linguistic information, attention-only decoder capacity can often replace FFN capacity for single-query grounding.

AI-use disclosure

AI tools assisted with language editing and software organization. The authors verified all methods, numbers, citations, figures, and conclusions.

References

[1] A. Kamath, M. Singh, Y. LeCun, G. Synnaeve, I. Misra, and N. Carion, “MDETR: Modulated detection for end-to-end multimodal understanding,” in Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 1780–1790.

[2] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su, J. Zhu, and L. Zhang, “Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection,” in European Conference on Computer Vision, 2024.

[3] S. Subramanian, W. Merrill, T. Darrell, M. Gardner, S. Singh, and A. Rohrbach, “ReCLIP: A strong zero-shot baseline for referring expression comprehension,” in Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, 2022, pp. 5198–5215.

[4] S. Kang, J. Kim, J. Kim, and S. J. Hwang, “Your large vision-language model only needs a few attention heads for visual grounding,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 9339–9350.

[5] S. Wu, S. Jin, W. Zhang, L. Xu, W. Liu, W. Li, and C. C. Loy, “F-lmm: Grounding frozen large multimodal models,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 24 710–24 721.

[6] H. Ndubuaku, K. Mosoyan, J. Mroz, N. Cylich, S. Kumar, P. Sandhu, R. Shemet, and J. H. Lee, “A controlled study of attention-only transformers,” 2026. [Online]. Available: https://arxiv.org/abs/2607.18363

[7] A. Sikarwar, A. Patel, and N. Goyal, “When can transformers ground and compose: Insights from compositional generalization benchmarks,” in Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, 2022, pp. 648–669.